Case Study 9: Smart Parking Fee Management

Aim: To develop a Python program that calculates parking charges based on vehicle category, entry time, exit time, overnight parking, daily limits, and penalty conditions while validating time inputs.

1. Problem Statement

A parking management system charges customers based on the duration of vehicle parking. Different vehicle categories have different hourly rates.

The program should:

1. Accept vehicle category.

2. Accept parking entry time and exit time.

3. Validate the entered time.

4. Calculate total parking duration.

5. Calculate parking fees according to vehicle category.

6. Identify overnight parking.

7. Apply daily maximum parking limits.

8. Apply additional penalties for invalid conditions.

9. Generate a final parking bill.

10. Assumed Parking Rules

| Vehicle category | Hourly rate |
|---|---|
| Two-wheeler | ₹20 |
| Four-wheeler | ₹50 |
| Heavy vehicle | ₹100 |

Additional rules:

- Maximum daily charge: ₹300 for two-wheelers.

- Maximum daily charge: ₹600 for four-wheelers.

- Maximum daily charge: ₹1,000 for heavy vehicles.

- Any part of an hour is charged as one full hour.
- Overnight parking is charged normally, with an additional ₹100 fee per night.
- Invalid time inputs are rejected.
- Exit time earlier than entry time is treated as next-day parking.

3. Algorithm

4. Start.

5. Enter vehicle category.

6. Enter entry time and exit time in HH:MM format.

7. Validate the time format.

8. Convert the time into minutes.

9. Calculate the parking duration.

10. If exit time is earlier than entry time, consider next-day parking.

11. Calculate the total parking hours.

12. Identify the vehicle category and hourly rate.

13. Calculate parking charges.

14. Apply the daily maximum limit.

15. Calculate overnight parking charges.

16. Display the final parking bill.

17. Stop.

18. Pseudocode

```
START

INPUT vehicle_category
INPUT entry_time
INPUT exit_time

VALIDATE entry_time AND exit_time
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IF time is invalid
    DISPLAY Invalid time
    STOP
END IF

CONVERT times into minutes

IF exit_time < entry_time
    duration = (1440 - entry_time) + exit_time
    overnight = TRUE
ELSE
    duration = exit_time - entry_time
    overnight = FALSE
END IF

CALCULATE parking hours
ROUND UP partial hours

SELECT hourly rate based on vehicle category

CALCULATE parking charge

APPLY maximum daily charge

IF overnight = TRUE
    ADD overnight fee
```

END IF

DISPLAY parking duration

DISPLAY parking charge

DISPLAY final bill

STOP

5. Python Program

Complete Python solution

# Smart Parking Fee Managementimport math# Vehicle parking ratesrates = { "two-wheeler": (20, 300), "four-wheeler": (50, 600), "heavy vehicle": (100, 1000)}# Function to validate and convert timedef convert_time(time): parts = time.split(":") if len(parts) != 2: return -1 if not parts[0].isdigit() or not parts[1].isdigit(): return -1 hour = int(parts[0]) minute = int(parts[1]) if hour < 0 or hour > 23 or minute < 0 or minute > 59: return -1 return hour * 60 + minute# Input vehicle categoryvehicle = input( "Enter vehicle category (two-wheeler/four-wheeler/heavy vehicle): ").lower()

6. Sample Output

Enter vehicle category: four-wheeler

Enter entry time (HH:MM): 20:00

Enter exit time (HH:MM): 08:00

====== PARKING BILL ======

Vehicle Category: four-wheeler

Entry Time: 20:00

Exit Time: 08:00

Parking Duration: 720 minutes

Parking Hours: 12

Hourly Rate: Rs. 50

Parking Fee: Rs. 600

Overnight Fee: Rs. 100

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Final Parking Fee: Rs. 700

====== THANK YOU ======

7. Flowchart

#chatgpt-mermaid-_r_2ch_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-

animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_2ch_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_2ch_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_2ch_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-mermaid-_r_2ch_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_2ch_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_2ch_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_2ch_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_2ch_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_2ch_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2ch_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2ch_ svg{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_2ch_ p{margin:0;}#chatgpt-mermaid-_r_2ch_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_2ch_ .label text,#chatgpt-mermaid-_r_2ch_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ .node rect,#chatgpt-mermaid-_r_2ch_ .node circle,#chatgpt-mermaid-_r_2ch_ .node ellipse,#chatgpt-mermaid-_r_2ch_ .node polygon,#chatgpt-mermaid-_r_2ch_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ .rough-node .label text,#chatgpt-mermaid-_r_2ch_ .node .label text,#chatgpt-mermaid-_r_2ch_ .image-shape .label,#chatgpt-mermaid-_r_2ch_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_2ch_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ .rough-node .label,#chatgpt-mermaid-_r_2ch_ .node .label,#chatgpt-mermaid-_r_2ch_ .image-shape .label,#chatgpt-mermaid-_r_2ch_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_2ch_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_2ch_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2ch_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2ch_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_2ch_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_2ch_ .edgeLabel p{background-color:rgb(0,

0, 0);}#chatgpt-mermaid-_r_2ch_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2ch_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_2ch_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_2ch_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2ch_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_2ch_ .icon-shape,#chatgpt-mermaid-_r_2ch_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_2ch_ .icon-shape p,#chatgpt-mermaid-_r_2ch_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_2ch_ .icon-shape .label rect,#chatgpt-mermaid-_r_2ch_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2ch_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_2ch_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_2ch_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_2ch_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_2ch_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_2ch_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_2ch_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].node .outer-path{filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_2ch_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_2ch_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2ch_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_2ch_-gradient);filter:drop-shadow( 1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2ch_ .node text{font-size:14px;font-weight:600;letter-spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_2ch_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_2ch_ .node tspan[font-weight="normal"],#chatgpt-mermaid-

_r_2ch_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_2ch_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ .node rect,#chatgpt-mermaid-_r_2ch_ .node circle,#chatgpt-mermaid-_r_2ch_ .node ellipse,#chatgpt-mermaid-_r_2ch_ .node polygon,#chatgpt-mermaid-_r_2ch_ .node path{fill:rgb(0, 40, 77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_2ch_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_2ch_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_2ch_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_2ch_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2ch_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter vehicle categoryEnter entry and exit timesValidate time inputsValid inputs?Display invalid inputConvert times into minutesExit earlier than entry?Calculate next-day durationCalculate same-day durationCalculate parking hoursSelect vehicle hourly rateCalculate parking feeApply daily maximum limitOvernight parking?Add overnight feeCalculate final billDisplay parking invoiceStopNoYesYesNoYesNo

8. Concepts Used from Syllabus

| Syllabus concept | Application |
| --- | --- |
| Variables | Entry time, exit time, parking fee |
| Data types | Integer, float, string |
| Operators | Arithmetic and comparison |
| Conditional statements | Vehicle and time validation |
| Loops | Can be extended for multiple vehicles |
| Strings | Time input and validation |
| Dictionaries | Vehicle parking rates |
| Tuples | Storing hourly and daily rates |

| | |
|---|---|
| Functions | Time conversion |
| Algorithms | Parking fee calculation |

## 9. Conclusion

The Smart Parking Fee Management program calculates parking charges based on vehicle category and parking duration. It validates time inputs, identifies overnight parking, applies daily maximum limits, and generates a parking invoice. This program demonstrates the practical application of Python programming in parking management systems.

Note: All parking rates and penalty amounts are assumed for this academic case study. This basic version handles one parking session crossing at most one midnight; a multi-day version would require processing each parking day separately.